

R5 — ML Research Track, Week 1

Member: _____

Project setup (do first)

1. Fork the repo: <https://github.com/slvDev/esp32-ai>
2. Clone your fork. Add the original as upstream.
3. Read README.md, RESULTS.md, and research/tinystories/README.md.
4. Write 5 lines in the shared Drive doc: what you think the project does and where the hard part is.

Your task this week

Goal: the team understands int4 PTQ before we quantize anything.

Steps: 1. Read research/tinystories/quantize_eval.py and src/quantize.py. 2. Write down what these terms mean: group size, fp16 scales, symmetric quantization, PTQ. 3. Write the 1-page PTQ explainer. Include the published table: baseline and ple at 4-bit.

Deliverable: the explainer as a markdown file in the club repo.

Verify: R2 can answer a 5-question quiz from the explainer.

Learning this week

Neural network basics. Watch 3Blue1Brown episode 3. Explain what a weight matrix does.

Resources

- quantize_eval.py: https://github.com/slvDev/esp32-ai/blob/main/research/tinystories/quantize_eval.py
- src/quantize.py: <https://github.com/slvDev/esp32-ai/blob/main/src/quantize.py>
- RESULTS.md (quantization section): <https://github.com/slvDev/esp32-ai/blob/main/RESULTS.md>
- 3Blue1Brown neural networks: https://www.youtube.com/playlist?list=PLZHQObOWTQDNU6R1_67000Dx_ZCJB-3pi

Done when

- [] PTQ explainer exists in the club repo
- [] R2 passes the 5-question quiz
- [] You can say what the project does in one sentence